



July 10, 2026

National Science Foundation

Small Business Innovation Research Program

2415 Eisenhower Avenue. Alexandria, VA 22314

Dear NSF SBIR Review Committee:

I am writing as an angel investor and Professor of Practice in Biomedical Engineering at Texas A&M. Across 30 years in product management and R&D at National Instruments and Maxim Integrated, I have mentored and backed entrepreneurs who brought hardware products to market. I write in strong support of ALTITUT LLC's VibeFab proposal led by PI Lu Sun.

The largest barrier for first-time hardware founders is the gap between creative intent and buildable design. A founder building a sensor wearable or IoT prototype must make dozens of decisions about component interfaces, voltage and current limitations, firmware, and fabrication specifications, each of which can kill a build if unresolved. No tool on the market helps non-expert creators catch these errors early.

General-purpose AI generates plausible instructions but does not check whether a design stays within datasheet ratings or flag missing specs. On the other hand, professional EDA platforms assume a trained engineer's fluency, which limits their use for early-stage startup founders who may lack this capability. VibeFab reasons the way an experienced hardware mentor does: it maps what the creator wants against what the components can do, surfaces the gaps, suggests bench tests, and tells the user when a qualified engineer needs to weigh in. That discipline, knowing when to stop, is what separates reliable design from guesswork.

If awarded, I will introduce VibeFab directly to the angel networks and early-stage funds in my network/portfolio, contribute an engineering review of its component reference



architectures, and deploy it as a standard tool in my own founder-mentoring practice to accelerate hardware startups toward investment readiness.

I strongly endorse PI Lu Sun and the ALTITUT LLC team. VibeFab addresses a real and costly gap in the hardware startup ecosystem.

Sincerely,

A handwritten signature in black ink that reads 'John Hanks'.

John Hanks

Angel Investor and Startup Mentor

Professor of Practice, Department of Biomedical Engineering, Texas A&M University

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